

配置实验项目运行环境

中国科学技术大学
编译原理课程组

- 使用 VirtualBox 配置 Ubuntu 22.04 系统
- 在 Ubuntu 22.04 系统中安装实验依赖软件

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- **VirtualBox**

- 免费和开源：开源软件，适用于教学。
- 跨平台性：支持多种操作系统，在任一系统上都可运行。
- 广泛虚拟机支持：可以运行各种不同的虚拟机系统。

- **VirtualBox安装**

- 官网下载链接：<https://www.virtualbox.org/>



• Ubuntu

- 免费和开源：免费的开源操作系统。
- 软件中心和包管理：提供方便的应用程序管理。
- 社区支持：拥有庞大的用户社区，支持长期维护。

• Ubuntu安装

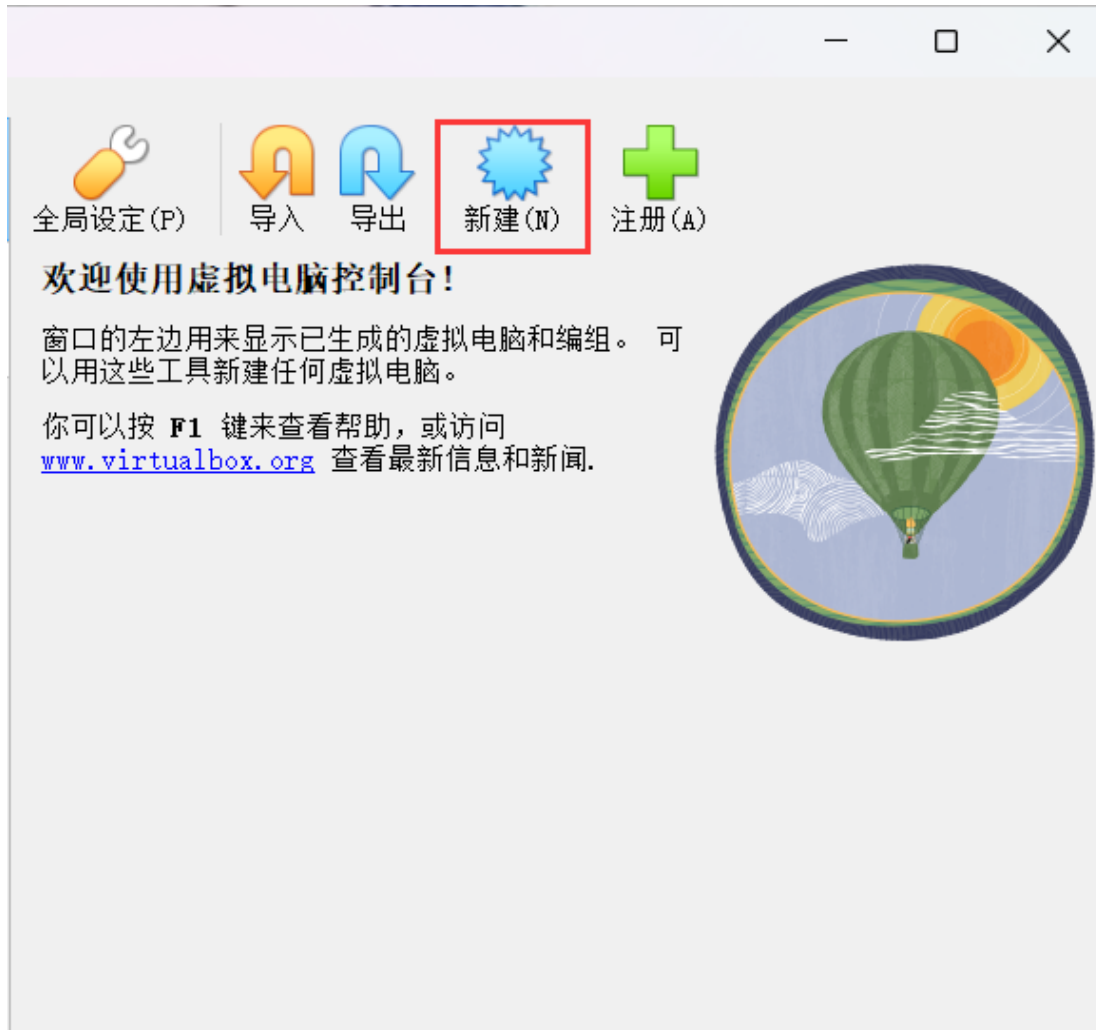
- 官网下载链接：
<https://releases.ubuntu.com/>
- 中科大镜像源下载链接：
<https://mirrors.ustc.edu.cn/ubuntu-releases/>

版本选择： ubuntu-22.04.3-live-server-amd64.iso

使用VirtualBox与系统文件搭建运行环境



• 步骤一：打开VB(VirtualBox)，点击新建：



使用VirtualBox与系统文件搭建运行环境



- **步骤二：在新建界面中设置名称与操作系统，之后点击下一步：**

使用VirtualBox与磁盘文件搭建运行环境



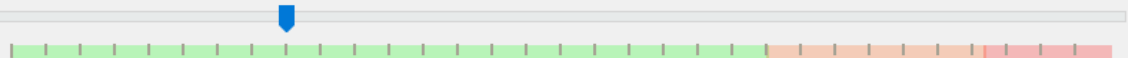
• 步骤三：设置虚拟机的内存大小与处理器数量：

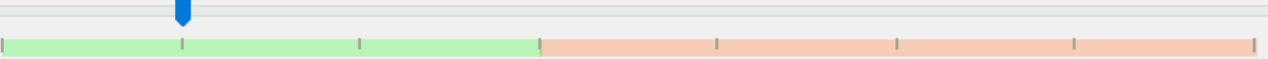
新建虚拟电脑



硬件

可设置虚拟电脑的内存大小和处理器数量。还可以启用EFI支持。

内存大小 (M) :  4096 MB

处理器 (P) :  2

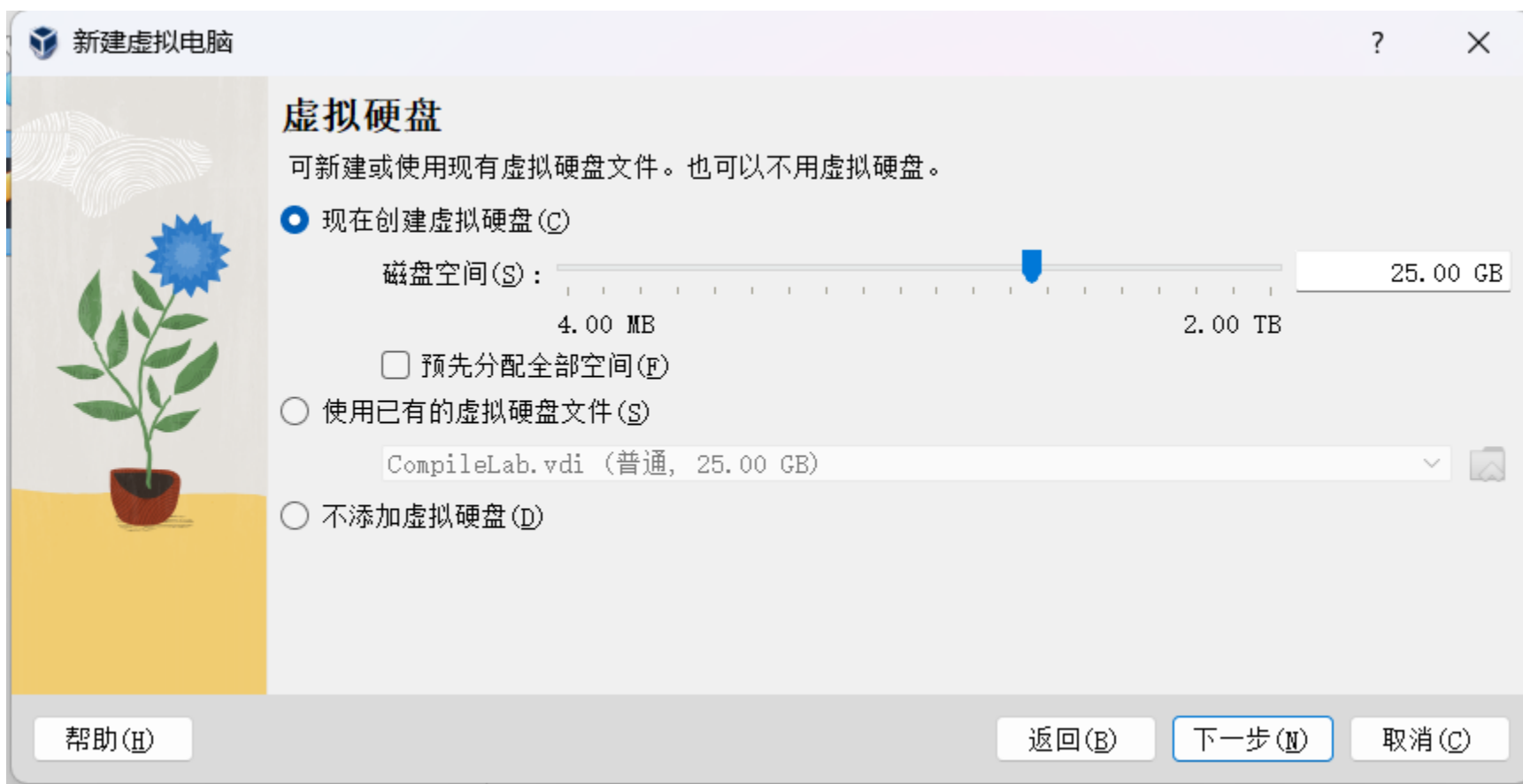
☐ 启用 EFI (只针对某些操作系统)

帮助 (H)返回 (B)下一步 (N)取消 (C)

使用VirtualBox与磁盘文件搭建运行环境



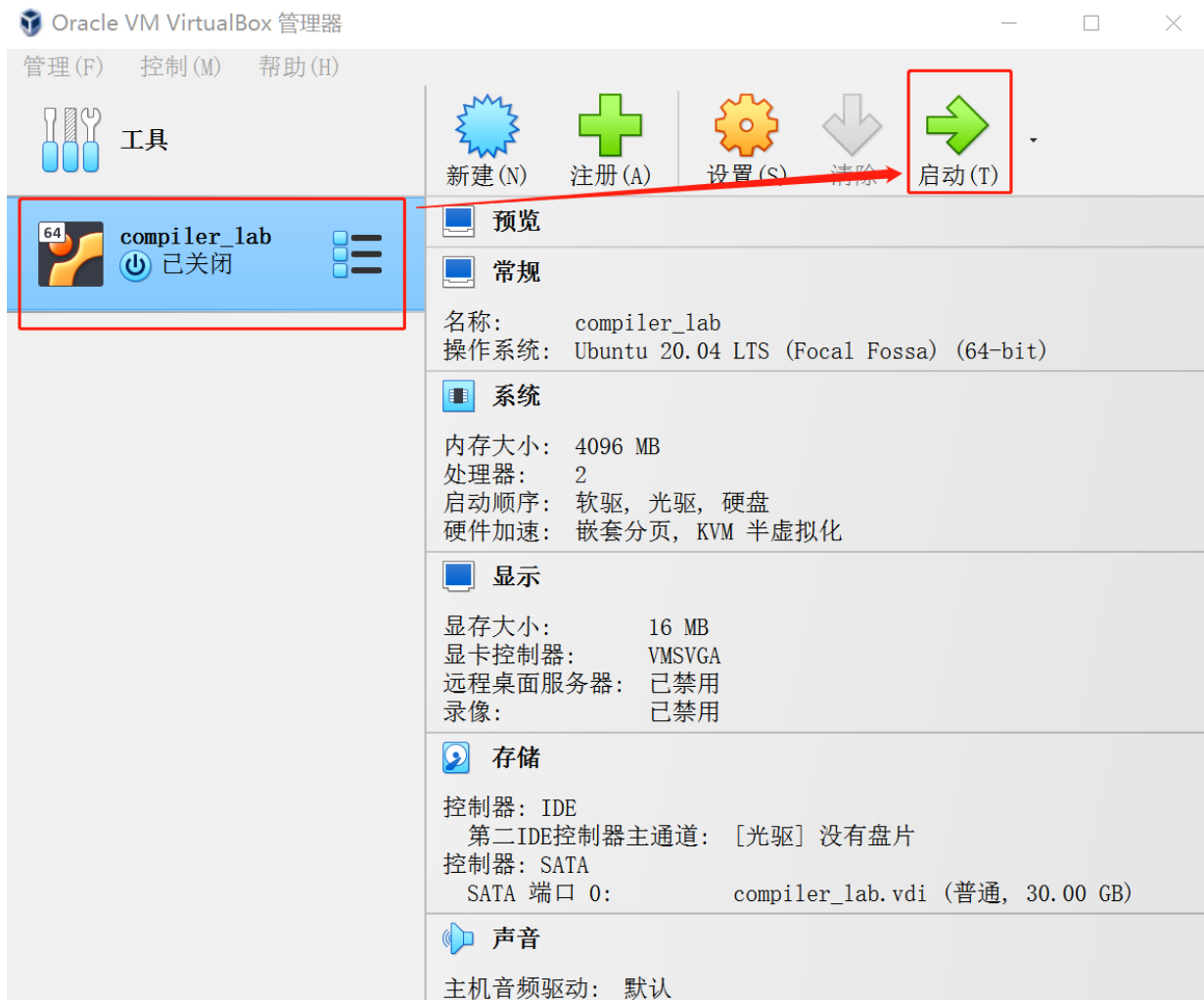
- **步骤四：接下来选择创建虚拟硬盘，设置最大磁盘大小为 25~40GB 即可：**



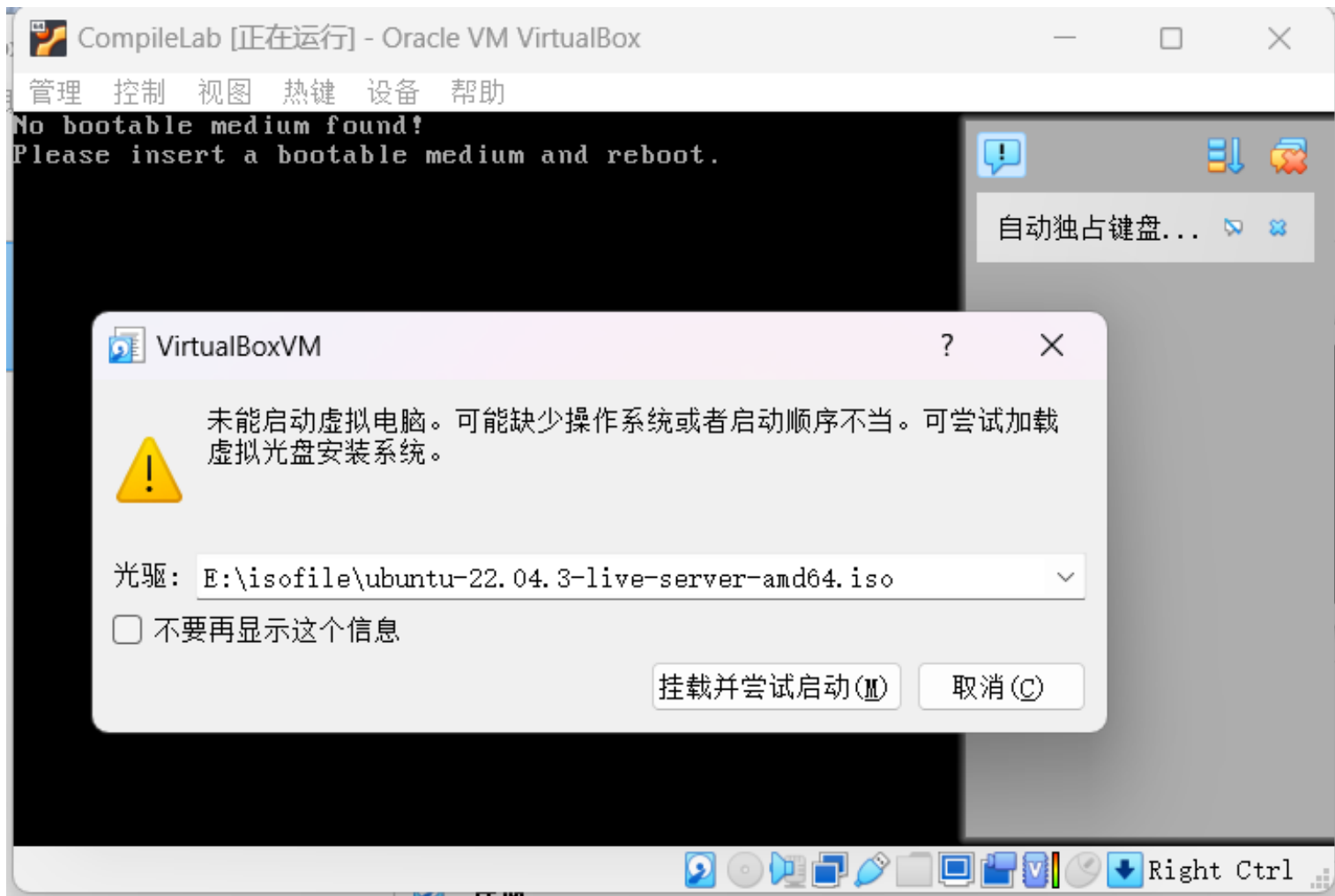
启动与登录虚拟机



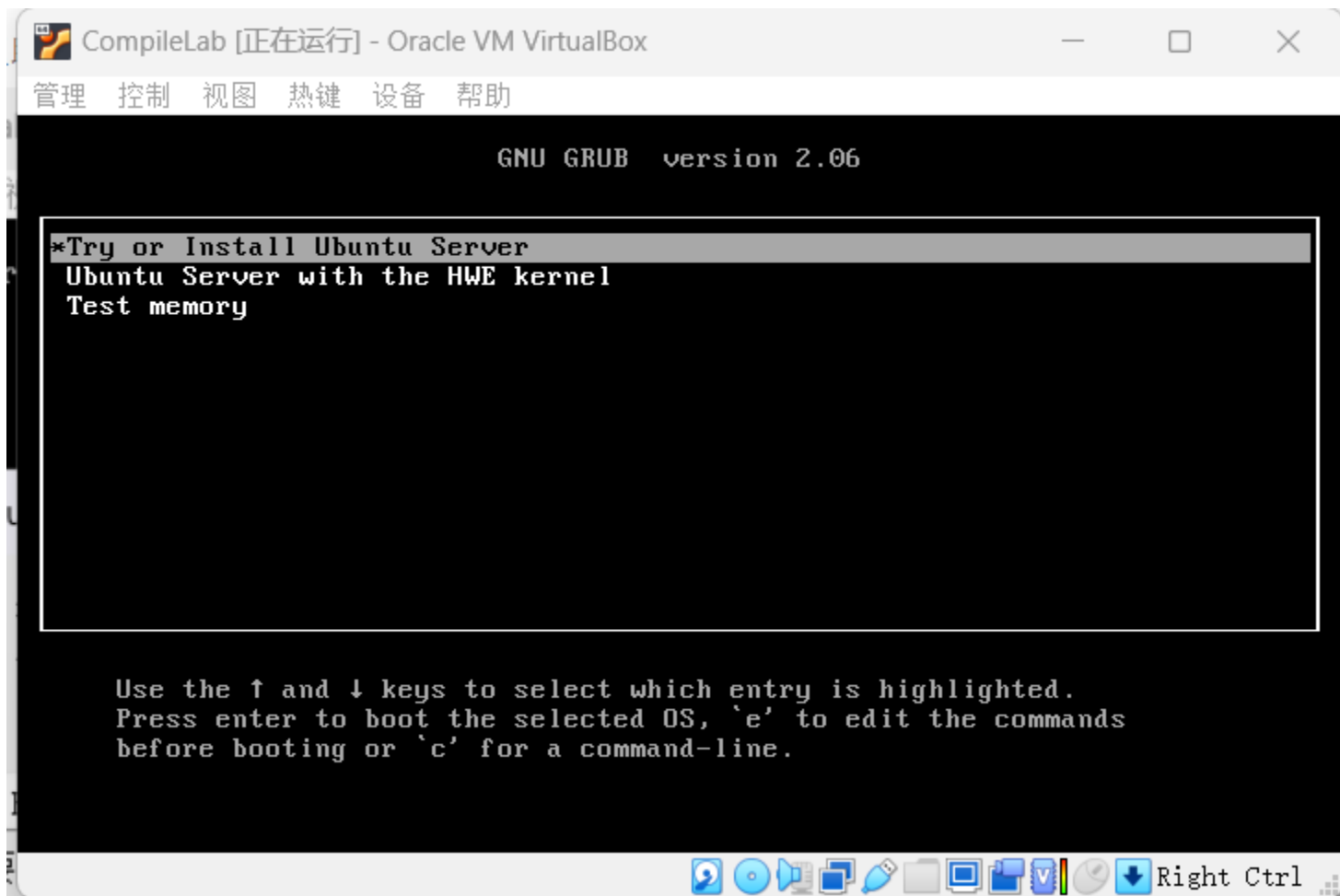
- **步骤五：虚拟机启动：选中刚刚创建的虚拟机，点击启动：**



• 步骤六：选中下载好的Ubuntu系统，并挂载并尝试启动：



- 步骤七：安装Ubuntu22.04系统，剩余操作选择默认后点击回车即可：



• 步骤八：注册虚拟机用户(用户名和密码需妥善保管):

Profile setup

[Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on the next screen but a password is still needed for sudo.

Your name:

Your server's name:
Server name must not be empty

Pick a username:
Username missing

Choose a password:
Password must be set

Confirm your password:

[Done]

- **步骤九：按照默认选择后点击回车直到成功安装系统后选择重启：**

```
Install complete! [ Help ]

curtin command curthooks
  configuring apt
  configuring apt
  installing missing packages
  configuring iscsi service
  configuring raid (mdadm) service
  installing kernel
  setting up swap
  apply networking config
  writing etc/fstab
  configuring multipath
  updating packages on target system
  configuring pollinate user-agent on target
  updating initramfs configuration
  configuring target system bootloader
  installing grub to target devices
final system configuration
  configuring cloud-init
  calculating extra packages to install
  installing openssh-server
    retrieving openssh-server
    curtin command system-install
    unpacking openssh-server
    curtin command system-install
  downloading and installing security updates
  curtin command in-target
  restoring apt configuration
  curtin command in-target
subiquity/Late/run

[ View full log ]
[ Reboot Now ]
```

- **步骤十：若出现界面卡顿，点击回车即可：**

```
[FAILED] Failed unmounting /cdrom.  
Please remove the installation medium, then press ENTER:  
[FAILED] Failed unmounting /cdrom.  
^[^A_
```

• 步骤十一：登录虚拟机,键入注册的用户名和密码:

```
ci-info: no authorized SSH keys fingerprints found for user student.
<14>Sep  6 09:09:19 cloud-init: #####
<14>Sep  6 09:09:19 cloud-init: -----BEGIN SSH HOST KEY FINGERPRINTS-----
<14>Sep  6 09:09:19 cloud-init: 1024 SHA256:2DeHt1ubsSdbFBgrK5g419udhgkEarWERCj00oI7uN4 root@compilelab (DSA)
<14>Sep  6 09:09:19 cloud-init: 256 SHA256:+u/5+v/DsuyJJ2AKOFU8XyY2NMYCe1CINrS05zMRxfY root@compilelab (ECDSA)
<14>Sep  6 09:09:19 cloud-init: 256 SHA256:YAxCyhc7LeH6qvWzaJe4rHDsJbQEVNP4uPdi0idjfr4 root@compilelab (ED25519)
<14>Sep  6 09:09:19 cloud-init: 3072 SHA256:p3p4PEXVxMqbdEqHlIrpz5Ud/WPspQQdT7r0CjpD5fkg root@compilelab (RSA)
<14>Sep  6 09:09:19 cloud-init: -----END SSH HOST KEY FINGERPRINTS-----
<14>Sep  6 09:09:19 cloud-init: #####
-----BEGIN SSH HOST KEY KEYS-----
ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBBCmJX/wMpxpMH8QP+B3a4kkHwoh0
PP4rWhfG6RY5KufVbb5mRzbz/TJ2z6wMwfGhurbWb/6r7X2dvW4sK2thLIY= root@compilelab
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIMsxuQ0AnsrUfo67pwx6ug3loUM/CRBe2IAk0viDRIGN root@compilelab
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGDULEWdv8aM6k0jm4IAw8pSQvYEudDWRxduj130VzhoqUN11Ff4QjL2FtAIzw9J
DMI/q7IjccM/CIOlnQ+031JSVFQjzE2ivsDwDv/tjIE4ppQNi5E2VjL9X81eFQjwIFdnqvJgail480PT3gN9G31LABFZ2C4q/Pal
VAJICUpf+szGf0H5iedddS3p4Jv1oStF0gFT0eC3/81112Ds0DDmeDvEupIZvjABvodFFDy2wskhk5V1okGltwuXecafy2B07++9
D4ltgbQzHbk66h1vIOUCupK//vxN9RSJVp8fXsJpmP18FOP8HbCfH77htfX1Uh847yc1Z++THpb0opzbKo/dAIjWyyhT2yV9PjU/
pKh+012M080E5k2y5n35T2FVsdRk0AFNus0Fa+I1mT7X1yu2FijuyCmgdKccB4fgJuuN63ocH2pgfESWgNayawJh8X0FMVpJBZ0
uHv7W0TZzN0U0Dn9jg16R7TM3jBwRySyl6qPG4b1TqKLtBox4eM= root@compilelab
-----END SSH HOST KEY KEYS-----
[ 27.946321] cloud-init[1355]: Cloud-init v. 23.2.1-0ubuntu0~22.04.1 finished at Wed, 06 Sep 2023
09:09:19 +0000. DataSource DataSourceNone. Up 27.94 seconds
[ 27.946783] cloud-init[1355]: 2023-09-06 09:09:19,969 - cc_final_message.py[WARNING]: Used fallback
ck datasource
compilelab login: student_
```


• 步骤十二：点击回车后成功登录虚拟机：

```
Ubuntu 22.04.3 LTS compilelab tty1

compilelab login:
compilelab login: student
Password:
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 5.15.0-83-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Wed Sep  6 12:35:00 PM UTC 2023

System load:  0.07861328125      Processes:           118
Usage of /:   44.3% of 11.21GB   Users logged in:     0
Memory usage: 5%                IPv4 address for enp0s3: 10.0.2.15
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

12 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Wed Sep  6 09:10:06 UTC 2023 on tty1
student@compilelab:~$
```

• 登录成功界面

```
compilerlab login: student
Password:
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.4.0-150-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Sat 10 Jun 2023 12:33:01 PM UTC

System load:  0.22               Processes:           111
Usage of /:   74.6% of 23.52GB   Users logged in:    0
Memory usage: 5%                IPv4 address for enp0s3: 10.0.2.15
Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

 * Introducing Expanded Security Maintenance for Applications.
   Receive updates to over 25,000 software packages with your
   Ubuntu Pro subscription. Free for personal use.

   https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

28 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Tue Jun  6 11:18:21 UTC 2023 from 10.0.2.2 on pts/0
student@compilerlab:~$ _
```

• 登录成功界面

```
compilerlab login: student
Password:
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.4.0-150-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Sat 10 Jun 2023 12:33:01 PM UTC

System load:  0.22               Processes:           111
Usage of /:   74.6% of 23.52GB   Users logged in:    0
Memory usage: 5%                IPv4 address for enp0s3: 10.0.2.15
Swap usage:   0%

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Last login: Tue Jun  6 11:18:21 UTC 2023 from 10.0.2.2 on pts/0
student@compilerlab:~$ _
```

成功安装 Ubuntu 后，接下来的我们会讲解在 Ubuntu 中安装实验依赖软件

- 使用 VirtualBox 配置 Ubuntu 22.04 系统
- 在 Ubuntu 22.04 系统中安装实验依赖软件

核心软件版本要求



- **实验使用的核心软件版本为**
 - Ubuntu 22.04
 - LLVM 14.0.0
 - Flex 2.6.4
 - Bison 3.8.2
- **本实验项目运行环境依赖于 Ubuntu 22.04 操作系统**
- **注：默认读者已掌握 Linux shell 的基本操作**

- **安装核心软件：**
 - build-essential, Flex, Bison, LLVM, Clang
- **安装的核心软件版本为：**
 - LLVM 14.0.0
 - Flex 2.6.4
 - Bison 3.8.2
- **下载依赖于 Ubuntu 22.04 操作系统**

- **安装命令：演示**

- 登录虚拟机后，在 terminal 中输入：
- `sudo apt update && sudo apt install build-essential flex bison llvm clang`

- **验证软件成功安装：演示**

- 登录虚拟机后，在 terminal 中输入：
- `gcc --version`, `flex --version`, `bison --version`, `llvm-as --version`, `clang --version`
- 如果出现对应软件的信息，即该软件安装成功

演示

- 登录虚拟机，在 terminal 中输入 `vim gcd.c` 创建测试文件 `gcd.c`
- 将以下代码输入至 `gcd.c` 中并保存

```
int gcd (int u, int v) {  
    if (v == 0) return u;  
    else return gcd(v, u - u / v * v);  
}  
int main() {  
    int x; int y; int temp;  
    x = 72;  
    y = 18;  
    if (x<y) {  
        temp = x;  
        x = y;  
        y = temp;  
    }  
    return gcd(x,y);  
}
```


演示

- 登录虚拟机，在 terminal 中输入 `vim gcd.c` 创建测试文件 `gcd.c`
- 将以下代码输入至 `gcd.c` 中并保存
- 在 terminal 中执行以下3条命令

```
clang -S -emit-llvm gcd.c
```

```
lli gcd.ll
```

```
echo $?
```

返回 18 则环境配置成功



下期再见！

Thanks!